## Table 6.2: notation of the leak codes

| code | meaning | evidence |
|---|---|---|
| M | identical variant row in training (same protein family, identical mutation set) | verified |
| m | every mutation in the row appears in training in the same protein, but not this combination (e.g. a test double whose two singles were trained) | verified |
| P | same protein (≥95 % identity over ≥90 % coverage, or ≥90 % identity over ≥50 %) in training, no mutation match | verified |
| H | strong homolog in training: MMseqs2 E ≤ 1e-3 over ≥50 % of the test protein | verified |
| h | weak/ambiguous: E ≤ 1e-2 or ≥30 % identity over ≥30 % of the test protein | measured |
| D | documented training on the same source dataset; rows not reconstructable | documented |
| C | pretraining-corpus exposure only (WT sequence/structure, no mutation supervision) | documented |
| U | training data not disclosed | documented |
| V | the dataset was used for validation or model selection | documented |
| # | the number is not a test result (the dataset is the training set) | verified |
| ? | probable but unverified (e.g. M?: same protein and same source dataset, row-level file not public) | inferred |
| n.a. | not audited: no statement about overlap is made, and the cell is not eligible for bold | – |

## Table 6.3: training inventories

| inventory | training data | source | evidence |
|---|---|---|---|
| ESM-MSR (released), ESM-MSR (devel), σ sweep, WT/MT heads | all substitutions of the 172 capped-train libraries (229,721 rows) | splits_oct06_capped.pkl + Tsuboyama2023 _Dataset2_Dataset3_20230416.csv | verified |
| MutateEverything st1 | the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background) | capped split; ME README recipe, stage 1 (20 epochs) | verified |
| MutateEverything st2 | the same singles plus all 53,188 measured doubles (207,546 rows); stage 2 (100 epochs) initialised from the same seed's stage 1 | capped split; ME README recipe | verified |
| ThermoMPNN (additive) | the same 154,358 singles | capped split; train_single.yaml | verified |
| ThermoMPNN-D | the 53,188 measured doubles plus 118,963 augmented doubles (upstream double-aug, built from the singles that have a Rosetta model: 79 % of them) | capped split; train_epistatic.yaml | verified |
| SPURS (retrained) | the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background) | capped split; patched SPURS training code, three seeds | verified |
| SPURS multi (retrained) | the same singles (through its same-seed SPURS initialisation) and the 53,188 measured doubles | capped split; patched SPURS training code, three seeds | verified |
| EsmTherm | K50_dG_Dataset1_Dataset2 domains; 414 wildtypes contribute train mutants under a per-row 80/10/10 split; only 47 domains are test-only | EsmTherm/datasets/split/wildtype_split. csv | protein-level (mutation rows unavailable) |
| IFUM | Megascale Dataset2: 322 train libraries, 86 in an unseparated val/test block | IFUM published dataset_splits.pdf | protein-level |
| DDGemb | S2450 = S2648 − 18 sequences/198 mutations >25 % identity to S669; 2,450 mutations / 115 UniProt | ddgemb.biocomp.unibo.it S2450.tsv/.fasta | verified |
| DDMut | S9028 = S2648 + FireProtDB, 4,514 forward + 4,514 reverse; multi-mutant branch SM1242 | biosig.lab.uq.edu.au S9028.csv, SM1242.csv | verified |
| GeoDDG-3D | S8754 = 8,754 single mutations / 301 proteins (ProThermDB + ThermoMutDB) | GeoStab S8754.csv | verified |
| ESM3ΔG | 960,216-sequence MGnify Stability fine-tune (525,127 WT + 435,089 point mutants); Megascale is its held-out test | Zenodo 19411306 mgnify_training_index.csv | verified (0 exact sequences) |
| ProSST | 18.8 M AlphaFoldDB structures (masked-LM); no ΔΔG fine-tune | ProSST paper | documented only |
| ProteinMPNN | PDB assemblies deposited before 2021-08-02; no mutation-level supervision | ProteinMPNN paper | documented only |
| ESM3 (small/medium/large, and the σ = 0 control) | UniRef/MGnify/JGI + PDB pre-2020-05-01 + AFDB v4 | ESM3 paper | documented only |
| Rosetta Cartesian ΔΔG | no training data | physics-based | |

## Table 6.5: what the audit finds

| finding | size | affected rows |
|---|---|---|
| 23 ProteinGym assays are fully inside the capped train split | 23/213 DMS, all Stability | 45,623/2,446,730 rows (100 % of those 23 DMS) |
| ME stage 1/2 and ThermoMPNN on those assays | singles (24,252 exact) + constituent (21,331) for ME; 24,252 exact for TMPNN | same 23 DMS |
| Domainome domain O95155_PF04564_1229 (UBE4B) is Megascale library 3L1X | 1 of 522 domains | 1,134 exact + 3,177 same-protein rows (0.8 % of Domainome) |
| EsmTherm's 80/10/10 is per row, not per domain: 27/33 test libraries contribute training mutants | 82.2 % of test rows same-protein | P, M? expected ≈ 0.8 |
| IFUM trained on 21/33 test libraries | 45.2 % of test rows same-protein | P, M? = same rows |
| ESM3ΔG is zero-shot on Megascale | 0 exact sequences anywhere in the benchmark | 40.7 % strong + 50.3 % weak homologs |
| DDGemb = S2450 | 92.5 % of S2648; 30 % of S8754; 28 % of Ssym; 45 % of Q3421; 26 % of K3822; 24 % of K2369 | upper bounds via S2648 |
| DDMut = S9028 + SM1242 | 88.8 % of S2648; 60.3 % of S8754; 42.1 % of S783; 19.6 % of S571; 29.7 % of Ssym; 58.9 % of PTMUL; 36.2 % of PTMUL-D | exact |
| GeoDDG = S8754 | S8754 is its training set (#); 97.7 % of S2648; 88.3 % of S783; 38.5 % of S571; 68.7 % of Q3421; 69.3 % of K3822; 76.1 % of K2369 | exact |
| the "six clean sets" claim | only S669 and S461 have zero identical mutations for every model | DDMut shares 11 %/16 % of them at protein level, GeoDDG 1 %/3 % |